Does It Matter Who Leads?

Principal Retirements and School Performance

October 2026

What is the impact of a change in school leadership?

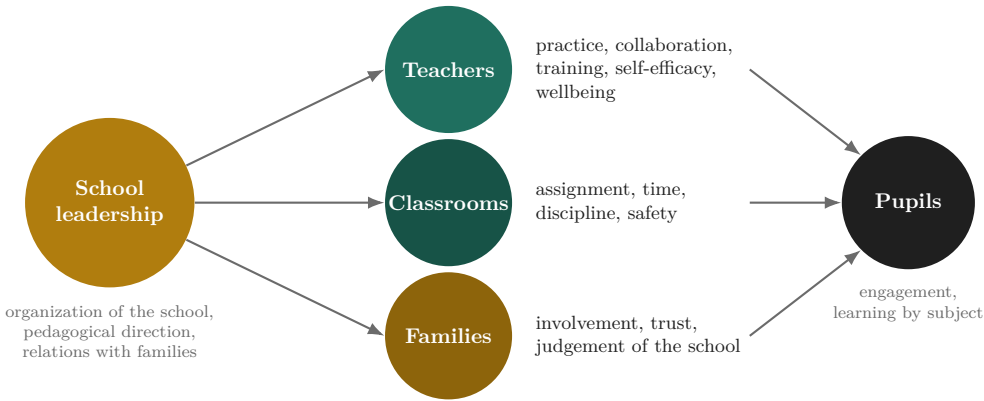
In a system where principals have limited authority and few teachers want the job.

Related literature

- One SD of principal quality moves achievement by 0.03 to 0.10 SD (Branch, Hanushek, and Rivkin, 2012; Bartanen, Husain, and Liebowitz, 2024).
- A change of principal lowers achievement by 0.03 SD and raises teacher exit (Bartanen, Grissom, and Rogers, 2019).
- The identified channel is personnel: who is hired, retained, and dismissed (Grissom and Bartanen, 2019; Muñoz and Prem, 2024).

What is not identified is the effect on the practice and effort of a fixed staff, on the organization of classrooms, and on parental inputs.

Causal map


School leadership
organization of the school, pedagogical direction, relations with families
Teachers
Classrooms
Families
practice, collaboration, training, self-efficacy, wellbeing
assignment, time, discipline, safety
involvement, trust, judgement of the school
Pupils
engagement, learning by subject

Institutional setting

- Spanish principals are teachers elected for four years. They do not hire, dismiss, set pay, or set the curriculum.
- A change of principal therefore changes leadership on a fixed staff.
- Civil-servant teachers may retire at 60 and must retire at 65, with extensions possible.

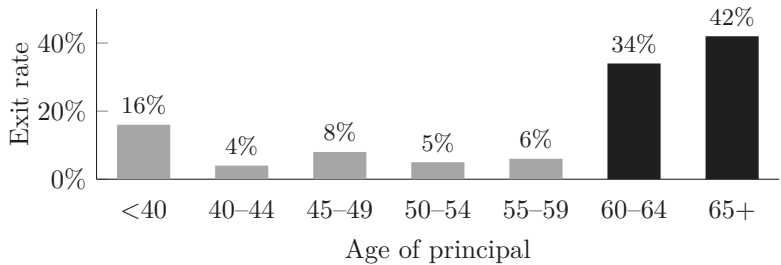
Data: LOMCE standardized tests in mathematics, Spanish, and English, with questionnaires to parents, teachers, and principals, for the universe of Madrid public primary schools, 2015–16 to 2018–19.

The selection of principals, Madrid 2015–2018

	Share
Competitions with a single candidate	75%
Competitions with two or more candidates	25%
Deserted competitions, filled by the administration	4%
Contested competitions not won by the top-merit candidate	39%

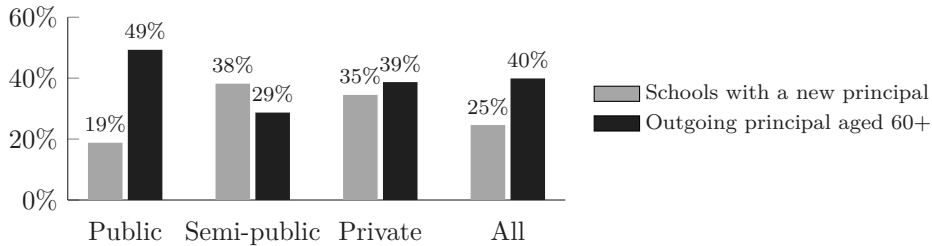
Source: official records of about 1,000 competitions, with candidates, merit scores, winners, and committees (Montalbán, 2026).

Principal exits by age



Note: Share of principals observed in one wave who are replaced by the next wave, by age in the last wave observed. A change is recorded when the respondent's sex or year of birth differs between consecutive waves (2016–17, 2017–18, 2018–19); a one-year difference in year of birth is treated as the same person. 1,884 principal-wave observations in 1,041 public schools. Dark bars: ages at or above 60, the early-retirement age for civil-servant teachers. *Source:* LOMCE principal questionnaires, Comunidad de Madrid.

Principal changes by school ownership



Note: 1,510 schools observed in two or more waves (1,041 public, 385 semi-public, 84 private); 401 principal changes (203, 167, 31). A change is recorded when the respondent's sex or year of birth differs between consecutive waves, 2016–17 to 2018–19, allowing a one-year difference in year of birth; 32 schools whose 2017 and 2019 respondents coincide but whose 2018 respondent differs are excluded. Ownership from the administrative school register. *Source:* LOMCE principal questionnaires, Comunidad de Madrid.

Identification strategy

- Instrument: the incumbent principal is aged 60 or older. The exit rate rises from about 6% to 34%.
- Treated cohorts: schools whose principal changes in 2018 or in 2019. Controls: schools not yet treated.
- Threat: retirement after 60 is voluntary, and may respond to how the school is doing.
- Test: among principals aged 60 or older, schools whose principal leaves and schools whose principal stays show the same levels and trends before the exit.

Empirical specification

Event study around the change of principal:

$$Y_{ist} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}[\, t - \tau_s = k \,] + \varepsilon_{ist}$$

Instrumental variables, with the age of the incumbent as instrument:

$$D_{st} = \pi \, \mathbf{1}[\text{age}_{s,t-1} \geq 60] + \alpha_s + \lambda_t + u_{st}, \qquad Y_{ist} = \beta \, \widehat{D}_{st} + \alpha_s + \lambda_t + \varepsilon_{ist}$$

τ_s: first school year under the new principal. i: a teacher, classroom, pupil, or parent.

The school is the unit of treatment; standard errors clustered by school; staggered-robust estimator.

Dispersion and heterogeneity of principal effects

Dispersion of principal effects, from excess variance around retirement-driven exits (Jones and Olken, 2005):

$$\text{Var}(\Delta Y_{st} \mid \text{exit}) - \text{Var}(\Delta Y_{st} \mid \text{no exit}) = 2\sigma_\theta^2$$

Heterogeneity by characteristics determined before the change:

$$Y_{ist} = \alpha_s + \lambda_t + \beta D_{st} + \delta\, D_{st} \times W_s + \varepsilon_{ist}$$

W_s includes how the vacancy was filled, the successor's merits and age relative to the incumbent, the successor's practices as a teacher in the previous year, and the staff's rating of the incumbent.

Challenges

- Selection into retirement: the age instrument is valid only if leavers and stayers over 60 are balanced before the exit.
- Power: about 100 retirement-driven changes over two cohorts. Enough for survey indices, tight for school-level test scores.
- Short windows: one or two years before and after each change.
- Measurement: changes are detected from the respondent's sex and year of birth; to be validated against appointment records.

Next steps

- Balance and pre-trends: leavers versus stayers among principals aged 60 or older.
- Power: within-school variance of each outcome and minimum detectable effects with 100 treated schools.
- Validation: changes in the questionnaires against named appointments in the competition records.

If the first test holds, the causal design stands. If not, the selection records support a descriptive paper on how principals are chosen.

Appendix: recent work on school and public-sector managers

Title	Journal	Year
Principal effects on teacher working conditions (Rahaman and Koedel)	CALDER WP	2026
Rethinking principal effects on student outcomes	J. Public Economics	2024
Managers' productivity and recruitment in the public sector: the case of school principals	AEJ: Economic Policy	2024
Personnel management and school productivity: evidence from India	Economic Journal	2024
The lasting impacts of middle school principals	NBER WP	2024
Managers in public schools (Tsao)	Princeton JMP	2024
Effects and mechanisms of CEOs quality in public education	Economic Journal	2023
School principal race, teacher racial diversity, and student achievement	J. Human Resources	2023
Managers and productivity in the public sector	Econometrica	2022

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Bartanen, B., A. N. Husain, and D. D. Liebowitz. 2024. "Rethinking Principal Effects on Student Outcomes." *Journal of Public Economics* 234.

Branch, G. F., E. A. Hanushek, and S. G. Rivkin. 2012. "Estimating the Effect of Leaders on Public Sector Productivity: The Case of School Principals." NBER Working Paper 17803.

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Jones, B. F., and B. A. Olken. 2005. "Do Leaders Matter? National Leadership and Growth Since World War II." *Quarterly Journal of Economics* 120(3): 835–864.

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Muñoz, P., and M. Prem. 2024. "Managers' Productivity and Recruitment in the Public Sector: The Case of School Principals." *American Economic Journal: Economic Policy* 16(4): 223–253.